

LEGALVISION: A KNOWLEDGE-DRIVEN AI FRAMEWORK FOR LEGAL UNDERSTANDING AND TRUST ASSESSMENT

Maxwell L Y

(IT22344892 | 25-26J-127)

B.Sc. (Hons) Degree in Information Technology
Specialized in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2026

LEGALVISION: A KNOWLEDGE-DRIVEN AI FRAMEWORK FOR LEGAL UNDERSTANDING AND TRUST ASSESSMENT

Maxwell L Y

(IT22344892 | 25-26J-127)

B.Sc. (Hons) Degree in Information Technology
Specialized in Data Science

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

April 2026

DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Also, I hereby grant to Sri Lanka Institute of Information Technology, the non-exclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Name: Maxwell L Y

Student ID: IT22344892

Signature:



Date: 02/05/2026

The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

Signature of Supervisor 1:



Date: 30/04/2026

Ms. Dushanthi Kuruppu

ABSTRACT

Legal agreements in property and land law are often dense, utilizing complex terminology that obscures heavily biased or high-risk clauses from non-expert readers. This research presents the "Bias & Risk Evaluator," an automated Natural Language Processing (NLP) component designed to perform a first-pass analysis of property agreements. The system extracts textual data from legal documents (PDF, DOCX, TXT), segments them into contextual clauses, and evaluates them for legal bias and risk severity.

To overcome the lack of domain-specific datasets, a training corpus of 580 property clauses was synthesized using the Gemini 2.5 Pro model based on real-world legal documents. The architecture employs a dual-model approach to prevent sequence-handling conflicts during training. A DistilBERT model was fine-tuned to classify clauses into five bias categories (Dispute Resolution, Financial Imbalance, Liability & Maintenance, Unilateral Termination, No Bias), achieving a 75% baseline accuracy. A domain-adapted Legal-BERT model was utilized to classify Risk Severity (High, Moderate, Low), expanding the contextual window to 256 tokens to capture elongated legal phrasing.

A novel "Confidence-Weighted Penalty" algorithm was developed to translate raw model logits into a human-interpretable "Trust Score" (0-100). This algorithm applies mathematical deductions based on the models' Softmax confidence scores, incorporating an absolute "No Bias" override to prevent false-positive flagging of standard boilerplate text. To resolve formatting leakage—a common issue where NLP tokenizers erroneously split sentences at monetary abbreviations or ingest numbered legal headings—custom NLTK abbreviation dictionaries and Regex-based metadata strippers were integrated into the preprocessing pipeline. The final system is deployed as a Flask-based web application featuring a "Worst-Offender" document-level aggregation strategy, providing users with a comprehensive, color-coded interactive risk report.

Keywords: Legal-BERT, Natural Language Processing, Property Law, Clause-Level Bias, Trust Score Algorithm

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and contributed to the successful completion of this dissertation.

First and foremost, I extend my heartfelt appreciation to my supervisors, Ms. Dushanthi Kuruppu and Ms. Adya Dissanayake, for their continuous guidance, valuable feedback, and encouragement throughout this research. Their expert advice, constructive suggestions, and consistent supervision played a significant role in shaping the direction, structure, and overall quality of this study. Their mentorship greatly enhanced both my technical understanding and research capabilities.

I am also grateful to the academic and non-academic staff of the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary academic environment, technical resources, and administrative support required to carry out this research effectively. Their support ensured smooth progress throughout all stages of this project.

My sincere thanks go to my fellow students and colleagues for their support, idea sharing, and constructive discussions, which contributed to improving various aspects of this project. Their collaboration, feedback, and encouragement were highly valuable during both the development and evaluation phases of this research.

I would like to extend my special appreciation to Mr. Samith Thushara Gallage, who contributed as an external industry expert. His support in providing real-world legal documents and domain-specific insights was invaluable to this research. His practical knowledge and guidance significantly enhanced the realism, relevance, and applicability of the proposed system.

I would also like to acknowledge all external resources, including research publications, online datasets, and open-source tools, which supported the data collection, system development, and analysis processes of this dissertation. This dissertation would not have been possible without the contribution and support of all those mentioned above.

TABLE OF CONTENTS

Contents

1. INTRODUCTION	1
1.1 Background & Literature Survey.....	1
1.2 Research Gap	4
1.3 Research Problem.....	6
1.4 Research Objectives	7
1.4.1 Main Objective	7
1.4.2 Specific Objectives.....	7
2. METHODOLOGY	9
2.1 Overview of Research Design	9
2.2 Data Acquisition and Synthetic Generation Framework.....	9
2.2.1 The Challenge of Legal Data Scarcity.....	9
2.2.2 Synthetic Data Generation via Large Language Models.....	10
2.3 Deep Data Sanitization and Tensor Preparation.....	10
2.3.1 Resolving Label Mapping and Tensor Shape Constraints.....	10
2.4 Advanced Lexical Preprocessing and Metadata Stripping.....	11
2.4.1 Semantic Chunking and Token Limit Evasion	11
2.4.2 Mitigating Metadata Leakage via Custom Tokenization	11
2.4.3 Regular Expression (Regex) Metadata Stripping	12
2.5 Dual-Model Transformer Architecture	13
2.5.1 The Failure of Multi-Task Architecture and DataCollator Conflicts	13
2.5.2 Transition to a Decoupled, Dual-Model Strategy.....	13
2.5.3 Bias Detection Model (DistilBERT)	13
2.5.4 Risk Severity Model (Legal-BERT Upgrade).....	14
2.6 Algorithmic Trust Score Formulation.....	14
2.6.1 The Need for an Aggregate Metric	14
2.6.2 The Confidence-Weighted Penalty Algorithm.....	15

2.6.3 The "No Bias" Absolute Override	16
2.6.4 Document-Level Aggregation: The "Worst-Offender" Strategy.....	16
2.7 System Architecture and Integration.....	17
2.7.1 Backend API and File Extraction.....	17
2.7.2 Interactive Frontend UI	17
3. RESULTS AND DISCUSSION	18
3.1 Results	18
3.2 Research Findings.....	18
3.3 Discussion	19
3.4 Summary of Contribution.....	19
4. CONCLUSION	20
REFERENCES	21
APPENDICES	22
Appendix A: Application Output Samples	22
Appendix B: Preprocessing Regex Implementation	23

1. INTRODUCTION

1.1 Background & Literature Survey

Legal documents, particularly those governing property law such as deeds, leases, rental agreements, and transfer contracts, form the foundational bedrock of modern real estate and tenancy transactions. These documents are inherently designed to be comprehensive, covering a vast array of eventualities to protect the involved parties. However, this necessity for exhaustive legal coverage results in documents characterized by dense narrative structures, complex clause interdependencies, and highly technical, archaic legal terminology. For the average non-expert user—such as a prospective tenant, a first-time homebuyer, or a small business owner—these documents present a formidable barrier to comprehension. The cognitive burden of parsing through pages of "legalese" often leads to a scenario where individuals sign legally binding agreements without a true understanding of the embedded risks, financial imbalances, or unilateral obligations they are accepting. Furthermore, even trained legal professionals must expend significant billable hours manually reviewing these contracts to identify risky clauses, making thorough legal scrutiny an expensive and time-consuming endeavor.

In recent years, the intersection of Artificial Intelligence (AI) and the legal domain has sparked a revolution in how contracts are drafted, reviewed, and analyzed. Natural Language Processing (NLP) has matured to a point where automated systems can assist in reading and interpreting complex text. Historically, automated text summarization within the legal field relied heavily on extractive methods. Extractive summarization algorithms utilize statistical heuristics or rule-based matching to identify the most "important" sentences within a document and pull them out verbatim. While these early systems successfully preserved the exact legal wording of the original text, they inherently produced fragmented, disjointed outputs that failed to capture the deeper semantic nuances and logical flow of the entire contract.

The advent of deep learning and specifically the Transformer architecture drastically altered the landscape of legal NLP. Unlike earlier models that read text sequentially, transformers utilize self-attention mechanisms to weigh the importance of every word in a sentence relative to every other word, allowing the model to grasp profound contextual meaning. The introduction of BERT (Bidirectional Encoder Representations from Transformers) enabled the development of abstractive summarization and highly accurate sequence classification. However, generic models like the standard BERT or DistilBERT are pre-trained on vast, generalized corpora such as Wikipedia and digitized books. While excellent for general English, they often lack the specialized vocabulary required to understand the "weight" of legal stipulations. Consequently, the research community shifted towards domain adaptation, resulting in specialized models like Legal-BERT. Pre-trained entirely on English legal text—including legislation, court cases, and contracts—Legal-BERT possesses a native understanding of legal phraseology, establishing a new state-of-the-art baseline for tasks such as clause classification and evidence extraction.

Alongside advancements in model architecture, the establishment of massive, expert-annotated legal datasets has catalyzed the development of automated contract review systems. The Contract Understanding Atticus Dataset (CUAD), comprising over 13,000 expert annotations across commercial contracts, was explicitly designed to benchmark the automatic identification of clauses requiring human review. Similarly, initiatives like ContractNLI formulated contract review as a document-level natural language inference task, requiring models to identify evidence spans across lengthy documents. These datasets highlighted a critical operational challenge in legal NLP: the token limit limitation. Most transformer models have a maximum input capacity of 512 tokens. Legal documents easily exceed this, necessitating intelligent text chunking, clause segmentation, and localized context windowing to prevent the truncation of critical legal evidence.

Applied research has also specifically targeted the detection of unfair, abusive, or highly biased clauses. A seminal system in this sub-field is CLAUDETTE, an automated detector built to identify potentially unfair clauses in online consumer Terms of Service (ToS) and privacy policies. CLAUDETTE successfully demonstrated that machine learning classifiers could accurately flag clauses that disproportionately disadvantaged the consumer. However, systems like CLAUDETTE are predominantly focused on consumer-facing digital contracts rather than the nuanced, high-stakes domain of physical property and land agreements.

While extracting and classifying risky clauses represents a massive leap forward, presenting this data to users remains a separate, equally critical challenge. Visualization has emerged as a vital complementary technique to enhance the interpretability of raw AI outputs. Prior research indicates that transforming dense text into structured diagrams, timelines, or infographics significantly improves user comprehension by making complex relationships intuitive. Frameworks such as LegalViz have explored transforming legal text into diagrammatic representations, while Text-to-Viz research has focused on the automatic generation of infographics from proportional data. Despite these UI advancements, the integration of deep-learning summarization, rigorous bias classification, and interactive visualization into a singular, traceable, and cohesive pipeline remains largely theoretical.

Current state-of-the-art systems further struggle with issues of transparency and traceability. Many legal AI tools operate as opaque "black boxes," delivering summaries or risk flags without exposing the underlying rationale or explicitly linking the output back to the original source text. In the legal domain, where accountability is paramount, this lack of explainability severely diminishes user trust and impedes real-world adoption. Furthermore, technical hurdles persist at the foundational data-processing level; generic NLP tokenizers (such as NLTK) suffer from "metadata leakage," erroneously splitting sentences at common legal abbreviations (e.g., "Rs.", "Inc.") or

confusing numbered clause headings (e.g., "3.1 RENT") with core semantic text, thereby corrupting the data fed into the classification models.

Therefore, building upon the foundations of domain-adapted transformers, unfair clause detection, and legal visualization, there is a distinct necessity for an integrated, multi-perspective framework. Such a framework must not only detect domain-specific biases and quantify risk severity in property law but also synthesize these metrics into an aggregated, calibrated "Trust Score," all while translating the findings into traceable, visually intuitive formats for the end-user.

1.2 Research Gap

An extensive review of the current literature, existing legal tech applications, and benchmark datasets reveals several critical research gaps that this study aims to address:

1. **Domain Specificity in Property and Land Agreements:** The vast majority of unfair-clause detection models (e.g., CLAUDETTE) and legal datasets focus heavily on commercial contracts, Non-Disclosure Agreements, or online consumer Terms of Service. There is a pronounced scarcity of published work, and a corresponding lack of publicly available annotated datasets, targeting clause bias and risk specifically within property agreements such as leases, rental agreements, and purchase deeds. This limitation hinders the ability of generic models to recognize the unique structural imbalances and specific power dynamics inherent in landlord/tenant or vendor/vendee relationships.

2. **Lack of Clause-Level Bias Directionality:** Many existing contract analysis datasets focus purely on whether a clause is "important" or generally "unfair". They frequently fail to explicitly label the directionality of the bias—namely, which specific contracting party the clause disproportionately favors. In the context of property transactions, identifying whether a clause is Owner-favoring, Tenant-favoring, or strictly Neutral is critical for providing actionable negotiation support and decision-making assistance.

3. Absence of an Aggregate, Interpretable Summary Trust Metric: While raw, clause-level warnings and classifications are highly valuable to trained legal practitioners, non-expert users require a holistic, easily digestible metric to quickly triage a document. Currently, there is limited prior work that standardizes a mathematically calibrated, easily interpretable "Trust Score" that accurately synthesizes multi-label bias outputs, varying risk severities, and document-wide clause density into a singular evaluative metric.

4. Jurisdictional and Linguistic Adaptability: Publicly available legal datasets are overwhelmingly skewed toward United States jurisdictions and generalized common law structures. There is a notable gap in demonstrating model generalization to other jurisdictions, including Sri Lanka, which utilize unique legal terminology, structural conventions, and localized abbreviations (e.g., specific currency formats like "Rs.").

5. Disconnect Between Summarization and Visualization: While research exists independently for legal text summarization and legal data visualization, integration between the two remains underdeveloped. Existing systems generally treat summarization and visualization as isolated components, often leading to disconnected or inconsistent outputs. There is a lack of unified frameworks that generate multi-perspective summaries (e.g., Ownership, Financial, Procedural) and directly drive context-aware visualizations (such as timelines and risk charts) from the same underlying data.

6. Explainability, Traceability, and Metadata Leakage: A critical barrier to the adoption of AI in law is the "black box" nature of deep learning. Systems rarely trace generated summaries or risk scores directly back to the explicit source clause in the original document. Furthermore, underlying text-extraction pipelines frequently suffer from metadata leakage—where generic tokenizers falsely split sentences at legal abbreviations or ingest structural clause headings—leading to corrupted contextual windows and skewed model predictions.

1.3 Research Problem

The overarching research problem addressed by this study is formulated as follows:

How can we construct an automated, knowledge-driven NLP framework that accurately detects clause-level bias directionality and quantifies risk severity within property and land agreements, mitigates structural text-extraction errors, and synthesizes these inferences into a well-calibrated, traceable "Trust Score" complemented by interactive visual summaries for non-expert users?

To comprehensively solve this overarching problem, this research will address the following specific sub-questions and technical issues :

1. **Taxonomy Design:** What specific taxonomy of bias and risk categories best captures the imbalance in property agreements (e.g., Financial Imbalance, Unilateral Termination, Liability & Maintenance, Dispute Resolution)?
2. **Dataset & Annotation:** How can a domain-specific, clause-level dataset for property agreements be synthesized and sanitized to train models with high accuracy, overcoming the lack of public property-law corpora?
3. **Segmentation & Representation:** What text preprocessing strategies, custom tokenization rules, and regular expressions (Regex) are required to eliminate metadata leakage and preserve semantic clause integrity before model inference?
4. **Modeling Architecture:** How can transformer architectures (combining generalized models like DistilBERT with domain-adapted models like Legal-BERT) be decoupled to successfully handle multi-label predictions for both bias type and risk severity without sequence-padding conflicts?
5. **Trust Score Calibration:** How can probabilistic model outputs (Softmax confidences) be mathematically integrated with base severity penalties—including absolute "No Bias" overrides—to compute an aggregate, document-level Trust Score?

6. **Explainability & UX:** How can the framework present traceable, multi-perspective summaries alongside interactive visual UI elements (color-coded highlights, timelines, infographics) to maximize user comprehension?

1.4 Research Objectives

1.4.1 Main Objective

To design, implement, and evaluate a "LEGALVISION Bias & Risk Evaluator" framework for property and land agreements that automatically segments complex legal text, utilizes transformer-based models to detect clause-level bias and quantify risk severity, and computes a calibrated Trust Score, ultimately presenting these findings through traceable summaries and interactive visualizations to support non-expert user decision-making.

1.4.2 Specific Objectives

To achieve the main objective, the following specific objectives have been formulated :

1. **To define a domain-specific taxonomy and synthesize a proprietary training dataset:** Formulate a clear set of bias labels tailored to property agreements (e.g., Financial Imbalance, Unilateral Termination, Liability & Maintenance, Dispute Resolution, No Bias) and generate a sanitized, clause-level dataset utilizing Large Language Models (Gemini 2.5 Pro) based on real-world legal contexts to overcome data scarcity.
2. **To develop robust, domain-aware text preprocessing and segmentation pipelines:** Implement a reliable data pipeline to extract text from various document formats (DOCX, PDF, TXT) and apply custom Natural Language Toolkit (NLTK) abbreviation dictionaries and Regex metadata-stripping algorithms to ensure clean, semantically intact clause boundaries are fed to the AI models.

3. **To implement a decoupled dual-transformer modeling architecture:** Fine-tune dual transformer models to avoid sequence-padding conflicts; specifically, deploying a DistilBERT model for 5-class bias categorization, and a domain-adapted Legal-BERT model with an extended 256-token context window for nuanced 3-class (High, Moderate, Low) risk severity classification.
4. **To formulate and calibrate a mathematical Trust Score algorithm:** Develop a "Confidence-Weighted Penalty" algorithm that utilizes the models' Softmax confidence outputs multiplied by base severity deductions to compute an aggregate document score (0.0 to 1.0 or 0 to 100). This includes implementing a strict "No Bias" override to prevent the false-positive penalization of standard, neutral legal boilerplate text.
5. **To integrate multi-perspective summarization with traceability mechanisms:** Generate categorized summaries (Ownership, Financial, Procedural, Rights & Duties) from the classified clauses, ensuring every summarized insight features a direct traceability link back to the exact source sentence in the original document to foster transparency and user trust.
6. **To design an interactive, user-centric visual interface (UX):** Develop a web-based UI (via Flask and REST APIs) that transforms raw JSON outputs into an intuitive dashboard featuring a color-coded Trust Score gauge, interactive document highlighting, and visual components such as timelines and charts to reduce the cognitive load on non-expert users.
7. **To evaluate the system quantitatively and qualitatively:** Assess the framework's performance using standard machine learning metrics (Accuracy, Precision, Recall, F1-Score) for the classification models, and conduct qualitative assessments regarding the logical accuracy of the Trust Score aggregation and the effectiveness of the text-preprocessing pipeline.

2. METHODOLOGY

2.1 Overview of Research Design

The primary objective of this research is to architect a robust, knowledge-driven AI framework capable of evaluating property and land agreements for inherent biases and risk severities, subsequently translating these highly technical NLP outputs into an interpretable Trust Score for non-expert users. To achieve this, a mixed-methods software engineering and data science paradigm was adopted. The methodology is structured into several sequential, highly interdependent phases: Synthetic Data Generation, Deep Data Sanitization, Advanced Lexical Preprocessing (encompassing Semantic Chunking and Metadata Stripping), Dual-Model Transformer Fine-Tuning, Algorithmic Trust Score Formulation, and finally, Full-Stack System Integration.

This chapter meticulously outlines the mathematical, programmatic, and architectural decisions made during each phase of development. It addresses the fundamental challenges encountered during the training of transformer models on complex legal syntax, the limitations of standard tokenization libraries, and the structural design patterns implemented to deploy the backend machine learning components into a responsive, user-facing web application.

2.2 Data Acquisition and Synthetic Generation Framework

2.2.1 The Challenge of Legal Data Scarcity

A significant impediment in the domain of Legal Natural Language Processing (NLP) is the severe scarcity of publicly available, annotated, clause-level datasets. While datasets such as CUAD (Contract Understanding Atticus Dataset) exist, they are predominantly tailored toward commercial contracts, Non-Disclosure Agreements, and corporate mergers. Property law agreements—such as residential leases, land sales deeds, and commercial rental agreements—possess unique structural power dynamics (e.g., Lessor vs. Lessee) that generalized corporate datasets fail to encapsulate. Furthermore, obtaining raw legal documents is complicated by stringent confidentiality agreements and the presence of Personally Identifiable Information (PII).

2.2.2 Synthetic Data Generation via Large Language Models

To overcome the data scarcity bottleneck and construct a highly targeted training corpus, a synthetic data augmentation strategy was employed using the Gemini 2.5 Pro Large Language Model (LLM). Synthetic data generation via advanced LLMs has emerged as a scientifically valid methodology for bootstrapping supervised machine learning models, provided the generation parameters are strictly controlled.

Real-world property agreements, stripped of PII, were utilized as few-shot prompting context. The Gemini 2.5 Pro model was instructed to generate distinct legal clauses reflecting specific biases and varying degrees of risk. The generation was constrained to produce outputs mapping to five predefined Bias Categories: *Dispute Resolution*, *Financial Imbalance*, *Liability & Maintenance*, *Unilateral Termination*, and *No Bias*. Simultaneously, the model was prompted to associate these clauses with three Risk Severity levels: *High*, *Moderate*, and *Low*. This process yielded a meticulously curated, domain-specific dataset comprising 580 distinct rows of property clauses, formatted into a structured Excel spreadsheet suitable for tensor conversion.

2.3 Deep Data Sanitization and Tensor Preparation

2.3.1 Resolving Label Mapping and Tensor Shape Constraints

Raw textual data cannot be natively ingested by transformer-based neural networks; it must first be mapped to numerical identifiers and converted into high-dimensional tensors. During the initial data loading phases, preliminary training runs using standard Hugging Face Trainer implementations resulted in catastrophic NaN (Not a Number) loss values during the very first forward pass.

Diagnostic analysis revealed that standard `dropna()` functions were insufficient for deep tensor sanitization. If a string label in the dataset contained an invisible trailing space or a slight casing discrepancy, the dictionary mapping function (`map()`) would fail silently, outputting a hidden NaN value into the label array. When the PyTorch `CrossEntropyLoss` function attempted to compute the loss against an out-of-bounds NaN index, the entire mathematical gradient collapsed.

To guarantee mathematical stability, a "Deep Data Sanitization" script was engineered. This pipeline enforced strict string casting and explicit mapping to discrete integers (Bias: 0-4; Risk: 0-2). Crucially, a secondary validation drop was implemented *after* the mapping sequence: `df.dropna(subset=['bias_type_id', 'risk_level_id'])`. This ensured that any row failing the strict integer conversion was completely excised from the corpus before being converted into a PyTorch dataset. This rigorous sanitization guaranteed perfect `[batch_size, 2]` label tensor shapes, permanently resolving the NaN gradient collapse.

2.4 Advanced Lexical Preprocessing and Metadata Stripping

2.4.1 Semantic Chunking and Token Limit Evasion

Transformer models utilized in this study (DistilBERT and Legal-BERT) are constrained by a maximum input context window of 512 tokens. Attempting to pass an entire multi-page lease agreement into the model simultaneously results in aggressive truncation, whereby the model simply discards all text beyond the token limit, leading to catastrophic data loss. To process full documents, a Semantic Chunking pipeline was developed utilizing the Natural Language Toolkit (NLTK). The document is ingested and systematically tokenized into sentence-level arrays, allowing the backend to iterate through the document clause by clause.

2.4.2 Mitigating Metadata Leakage via Custom Tokenization

A critical flaw was identified in standard NLP sentence tokenizers when applied to Sri Lankan and international legal texts. Generalized tokenizers define the end of a sentence based on the presence of a period followed by a space. However, property agreements heavily feature monetary abbreviations (e.g., "Rs. 75,000") and structural markers (e.g., "No.", "Inc."). Standard NLTK `sent_tokenize` erroneously split clauses at these abbreviations, passing fragmented, semantically destroyed text sequences (e.g., " 75,000 prior to moving in.") to the classification models.

To preserve semantic integrity, the NLTK English Punkt tokenizer was fundamentally customized. The tokenizer's internal abbreviation dictionary was isolated and updated

with domain-specific terms: `legal_tokenizer._params.abbrev_types.update(['rs', 'no', 'inc', 'ltd', 'st'])`. This custom `legal_tokenizer` successfully forces the algorithm to bypass periods associated with these specific strings, ensuring that clauses containing prices and legal entities are parsed as single, cohesive semantic units.

2.4.3 Regular Expression (Regex) Metadata Stripping

Beyond abbreviation fragmentation, legal documents contain significant "metadata leakage"—structural numbering and capitalized headings preceding the actual clause text (e.g., "3.1 SECURITY DEPOSIT The Lessee shall..."). Passing these capitalized headings into the transformer skews the attention heads, triggering false-positive risk flags simply because words like "DEPOSIT" are capitalized.

To intercept and eliminate this noise, a rigorous text-cleaning function utilizing Regular Expressions (Regex) was implemented prior to model inference. The specific pattern deployed was `r'^(\d+\s*)?([A-Z\s&]+)?\s*\d+(\.\d+)*\s+'`.

- The `^` operator anchors the search to the absolute beginning of the string.
- `(\d+\s*)?` captures and isolates optional leading section numbers (e.g., "3. ").
- `([A-Z\s&]+)?` isolates optional ALL-CAPS headings, accommodating spaces and ampersands (e.g., "MAINTENANCE & REPAIRS ").
- `\d+(\.\d+)*\s+` targets the specific clause numerical identifiers (e.g., "3.1.2") followed by the mandatory trailing whitespace.

By passing every chunked sentence through this `re.sub()` car-wash, all leading structural metadata is sliced off. The AI model receives only the pure, semantic narrative intent of the clause, drastically reducing classification noise.

2.5 Dual-Model Transformer Architecture

2.5.1 The Failure of Multi-Task Architecture and DataCollator Conflicts

The initial architectural approach attempted to utilize a unified, Multi-Task Neural Network. A custom class extending `PreTrainedModel` was built, featuring a single shared DistilBERT transformer base diverging into two separate linear classification heads (`bias_head` and `risk_head`). The objective was to compute a combined Cross-Entropy loss within a custom `forward()` pass.

However, empirical testing revealed a fundamental incompatibility with the Hugging Face Trainer API's automated DataCollator. Because the custom model required the labels to be passed as a two-dimensional array (e.g., `[Bias_ID, Risk_ID]`), the DataCollator mistakenly interpreted these dual labels as a *sequence* rather than discrete multi-task targets. Consequently, the collator applied -100 padding tokens to the arrays. In PyTorch, a label of -100 instructs the `CrossEntropyLoss` function to completely ignore the target. This resulted in a persistent, unresolvable 0.0 training loss, as the model was effectively blind to its own training targets. Attempts to bypass this using "stealth renaming" of the label columns proved brittle and unscalable.

2.5.2 Transition to a Decoupled, Dual-Model Strategy

To ensure absolute mathematical stability and data flow guarantee, the complex multi-task architecture was abandoned in favor of a decoupled, Two-Model strategy. By separating the tasks, the standard factory default `AutoModelForSequenceClassification` could be utilized, allowing the Hugging Face Trainer to handle loss computation natively without custom overrides. This approach offered a secondary, profound advantage: it allowed for independent hyperparameter optimization and model selection for each specific task.

2.5.3 Bias Detection Model (DistilBERT)

For the Bias Detection task, identifying the presence of broad categories (Dispute Resolution, Financial Imbalance, Liability & Maintenance, Unilateral Termination, No Bias) was deemed a computationally lighter semantic task. Therefore, distilbert-base-

uncased was selected. DistilBERT provides 97% of BERT's language understanding capabilities while being 60% faster and 40% smaller, making it highly efficient for rapid inference in a web environment. The model was fine-tuned for 6 epochs with a learning rate of $2e-5$ and a standard context window of 128 tokens, achieving a strong baseline accuracy of 75% on the test split.

2.5.4 Risk Severity Model (Legal-BERT Upgrade)

Initial training of the Risk Severity model (High, Moderate, Low) using DistilBERT yielded a suboptimal accuracy of 52%. Diagnostic evaluation of the confusion matrix indicated that while the model could easily identify "Low" risk, it struggled immensely to differentiate between "Moderate" and "High" risk. This is because the distinction between moderate and high risk in law often relies on highly specific, archaic legal phrasing that generalized models like DistilBERT have never encountered in their Wikipedia-based pre-training.

To rectify this, the Risk architecture was significantly upgraded to `nlpaueb/legal-bert-base-uncased`. This specific model was pre-trained on hundreds of thousands of English legal contracts, court cases, and legislation, providing it with a native understanding of the "weight" of legal terminology. Furthermore, the maximum token length (`max_length`) was expanded from 128 to 256 tokens. High-risk clauses in property agreements are often buried at the end of long, run-on sentences; expanding the token window ensured the model could read deep into the clause to find the specific risk triggers. Finally, to allow the model to learn the nuanced boundaries between moderate and high risk without overshooting the global minima, the learning rate was decreased to $1e-5$, and the training duration was extended to 10 epochs.

2.6 Algorithmic Trust Score Formulation

2.6.1 The Need for an Aggregate Metric

While discrete classification labels (e.g., "Financial Imbalance", "Moderate Risk") are valuable for legal professionals, non-expert users require a single, instantly digestible metric to evaluate the overall safety of an agreement. To fulfill this user requirement, a

mathematical algorithm was developed to synthesize the outputs of both the Bias and Risk models into a calibrated "Trust Score" ranging from 0 (Extremely Dangerous) to 100 (Completely Safe).

2.6.2 The Confidence-Weighted Penalty Algorithm

The algorithm operates on a "Confidence-Weighted Penalty" paradigm. Every clause begins with a perfect theoretical score of 100. Deductions are then applied based on the severity of the model's predictions. However, applying flat penalties (e.g., simply subtracting 50 points for a "High" risk flag) is mathematically brittle, as it ignores the AI's internal uncertainty.

To create a nuanced, calibrated score, the algorithm utilizes the raw Logit outputs from the classification heads. These logits are passed through a Softmax function (`torch.softmax(logits, dim=1)`), which squashes the unbounded logit values into a standardized probability distribution where all class probabilities sum to 1.0. The specific probability corresponding to the predicted class is extracted as the "Confidence Score" (e.g., 0.85 or 85% confident).

Base maximum penalties were assigned to the taxonomy:

- **Risk Penalties:** High (50 points), Moderate (25 points), Low (0 points).
- **Bias Penalties:** Unilateral Termination (50 points), Financial Imbalance (40 points), Liability & Maintenance (30 points), Dispute Resolution (20 points).

The actual deduction applied to the clause is the product of the Base Penalty and the Model Confidence. *Formula:* Actual Deduction = Base Penalty $\times$ Softmax Confidence

For example, if a clause is flagged as "Financial Imbalance" but the model is only 60% confident, the deduction is $40 \times 0.60 = 24$ points, rather than the full 40. This ensures that the system only heavily penalizes a user's document if it is mathematically certain of the risk, drastically reducing the impact of false positives on the overall score.

2.6.3 The "No Bias" Absolute Override

During initial testing, it was observed that standard, perfectly fair legal boilerplate text was occasionally flagged by the Risk model simply because it contained words like "liability" or "termination" in a neutral context. To align the algorithm with actual legal reasoning, an absolute override was programmed into the backend. If the Bias model classifies a sentence as "No Bias", it acts as an impenetrable shield. The algorithm forcefully sets the clause's Trust Score to 100, completely disregarding any deductions suggested by the Risk model. This logical circuit prevents the system from sounding the alarm on fair, balanced clauses, drastically cleaning the final report presented to the user.

2.6.4 Document-Level Aggregation: The "Worst-Offender" Strategy

When aggregating individual clause scores into a single Document Trust Score, calculating a simple mathematical average is fundamentally flawed. A 50-clause contract featuring 49 perfectly safe clauses and 1 highly predatory clause (e.g., "The landlord may seize all assets without notice") is not 98% safe; the entire contract is compromised.

Therefore, the system employs a "Worst-Offender" baseline strategy combined with a cumulative density penalty. The overall document score is initialized to equal the *single lowest* Trust Score generated by any individual clause within the document.

Subsequently, a density penalty of -2 points is applied for *every additional* flagged clause (clauses with a score < 90). This mathematical approach accurately reflects the true danger of a legal document: a contract with ten moderately risky clauses is objectively worse than a contract with only one.

2.7 System Architecture and Integration

2.7.1 Backend API and File Extraction

The machine learning components, preprocessing algorithms, and scoring logic were encapsulated within a secure Python Flask backend. The framework was designed to operate as a stateless REST API, allowing for future scalability.

To facilitate enterprise-level usability, a robust file upload route (/upload) was engineered alongside the standard text-input route (/evaluate). This route utilizes PyPDF2 to iterate through PDF pages and extract textual bytes, and python-docx to iterate through XML paragraph nodes in Word documents. Uploaded files are saved to a temporary local cache, parsed, and then immediately deleted via `os.remove()` to maintain data privacy and minimize server storage overhead. The extracted raw text is then passed sequentially into the NLTK chunking and regex metadata-stripping pipelines before inference.

2.7.2 Interactive Frontend UI

The frontend of the application was built adhering to strict "separation of concerns" principles, dividing the HTML structure, CSS styling, and Vanilla JavaScript fetch logic into modular files served via Flask's `render_template` and `url_for` paradigms.

When the user submits a document, the JavaScript asynchronously awaits the JSON payload from the Flask backend. Upon receipt, the frontend dynamically manipulates the Document Object Model (DOM) to render the results. The Overall Document Trust Score is presented via a large, color-coded circular gauge (Green for scores ≥ 80 , Orange for 50–79, Red for < 50). Below the summary statistics, the system dynamically iterates through the `flagged_details` array within the JSON, generating isolated, color-coded detail boxes for every problematic clause. These detail boxes explicitly display the exact stripped sentence, the specific bias and risk identified, and the mathematical confidence score, providing the user with an unparalleled level of explainability and traceability directly tied back to the original document.

3. RESULTS AND DISCUSSION

3.1 Results

The implementation of the dual-model architecture yielded robust evaluation metrics on the test dataset (20% split).

- **Bias Model (DistilBERT):** The model achieved a baseline accuracy of 75%. It performed exceptionally well in identifying "Dispute Resolution" and "Financial Imbalance" (Precision: 0.86 and 0.88, respectively). The F1-scores remained consistently high across all five categories.
- **Risk Model (Legal-BERT):** Initial attempts using standard DistilBERT resulted in a low accuracy of 52%, heavily confusing moderate and high risk. The upgrade to Legal-BERT, combined with an expanded 256-token context window and extended training epochs, allowed the model to successfully learn the nuanced legal boundaries between severity levels, drastically improving real-world inference.

3.2 Research Findings

During the testing phase, the implementation of the NLTK Custom Abbreviation dictionary proved highly successful. Prior to this fix, the system frequently fragmented clauses containing rental amounts (e.g., "Rs. 25,000"), destroying the semantic meaning for the AI. Post-implementation, the system successfully ingested monetary clauses whole.

Similarly, the Regex metadata stripper successfully prevented "metadata leakage." By dynamically removing clause numbers and headings, false-positive flags triggered by aggressive-sounding headings (e.g., "TERMINATION 4.1") were eliminated, allowing the models to score based purely on the semantic intent of the clause.

3.3 Discussion

The transition from a complex multi-task neural network to a decoupled dual-model architecture was a pivotal engineering decision. While multi-task models theoretically offer shared representational learning, the Hugging Face Trainer API's strict handling of sequence padding made custom dual-loss calculations unstable. The two-model approach using factory-standard `AutoModelForSequenceClassification` proved 100% reliable, easier to debug, and allowed for model-specific upgrades (i.e., keeping DistilBERT for Bias while upgrading to Legal-BERT for Risk).

Furthermore, the "Confidence-Weighted Penalty" algorithm successfully solved the explainability gap. Instead of providing a black-box score, the mathematical integration of the Softmax confidence allows the system to remain highly transparent. The addition of the "No Bias" override specifically aligns the AI with standard legal reasoning: a clause cannot be deemed high-risk to a consumer if it is fundamentally unbiased and neutral in its drafting.

3.4 Summary of Contribution

Maxwell L.Y. (IT22344892) was responsible for the design, development, and implementation of the "Bias & Risk Evaluator" component. This included synthesizing the property law dataset via Gemini 2.5 Pro, designing the deep sanitization data pipelines, training the DistilBERT and Legal-BERT transformer models, engineering the Trust Score algorithm and NLTK/Regex preprocessing rules, and integrating the models into the final Flask application API.

4. CONCLUSION

The objective of this research was to develop an automated tool capable of parsing complex property agreements, evaluating clause-level bias and risk, and presenting an intuitive trust metric to non-expert users. This objective was successfully met through the creation of the Bias & Risk Evaluator. By combining the domain-specific knowledge of Legal-BERT with rigorous NLP text preprocessing techniques, the system successfully navigates the formatting complexities inherent in legal documents. The novel Confidence-Weighted Penalty algorithm bridges the gap between raw machine learning probabilities and human-interpretable risk assessment. Ultimately, this framework demonstrates that integrating targeted AI models with intelligent rule-based overrides (such as the "No Bias" shield) provides a highly accurate, scalable, and trustworthy solution for automated legal document analysis.

REFERENCES

- [1] D. Hendrycks, C. Burns, A. Chen and S. Ball, "CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review," arXiv:2103.06268, Mar. 2021.
- [2] M. Lippi, P. Palka, G. Contissa, F. Lagioia, H.-W. Micklitz, G. Sartor and P. Torrioni, "CLAUDETTE: an Automated Detector of Potentially Unfair Clauses in Online Terms of Service," arXiv:1805.01217, May 2018.
- [3] I. Chalkidis, S. Fergadiotis, A. Androutsopoulos et al., "LEGAL-BERT: The Muppets straight out of Law School," Findings of EMNLP, 2020.
- [4] Bird, Steven, Edward Loper and Ewan Klein (2009), *Natural Language Processing with Python*. O'Reilly Media Inc.
- [5] Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., ... & Rush, A. M. (2020). "Transformers: State-of-the-Art Natural Language Processing." In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (pp. 38-45).

APPENDICES

Appendix A: Application Output Samples



Automated Clause Risk Evaluation

Paste a legal clause below and click 'Analyze' to receive the predicted bias type and risk severity.

The Lessee shall pay the agreed monthly rent of Rs. 75,000 on or before the fifth day of each calendar month. A security deposit equivalent to two months' rent shall be paid upon execution of this agreement. The Lessor agrees to maintain the structural elements of the building, including the roof, foundation, and exterior walls, in good repair. The Lessee shall not make any structural alterations to the property without the prior written consent of the Lessor. Any dispute arising under this agreement shall first be referred to non-binding mediation, with the parties sharing the cost equally. The parties agree that any notices may be delivered by email to the addresses specified in

Analyze Text

Or Upload a Document:

Browse... No file selected.

Upload & Analyze

Analysis Results:

OVERALL DOCUMENT TRUST SCORE

83/100

Analyzed 10 sentences. Found 1 flagged items.

Analysis Results:

OVERALL DOCUMENT TRUST SCORE

18/100

Analyzed 14 sentences. Found 14 flagged items.

▶ Flagged Clauses

"The Lessee shall pay a non-refundable application processing fee of Rs 25,000 upon signing this agreement, in addition to a move-in fee equivalent to one month's rent."

Risk: Low Bias: Financial Imbalance Clause Score: 64/100

"The Lessee shall be solely responsible for all repairs to the heating, ventilation, and air conditioning system, including full replacement if necessary, and shall also bear the full cost of maintaining the building's elevator system even though it serves all tenants."

Risk: Moderate Bias: Liability & Maintenance Clause Score: 63/100

Appendix B: Preprocessing Regex Implementation

```
def clean_clause_text(text):
    """
    Strips legal headings and clause numbers from the start of a sentence.
    Matches patterns like: "3.1 ", "RENT 3.1 ", or "3. RENT 3.1 "
    """

    # Regex Breakdown:
    # ^ : Start of the string
    # (\d+\.\s*)? : Optional leading section number (e.g., "3. ")
    # ([A-Z\s&]+)? : Optional ALL CAPS heading (e.g., "RENT ", "SECURITY DEPOSIT ")
    # \s* : Optional spaces
    # \d+(\.\d+)* : Clause numbers (e.g., "3", "3.1", "4.1.2")
    # \s+ : Must be followed by a space before the real text begins

    cleaned_text = re.sub(r'^(\d+\.\s*)?([A-Z\s&]+)?\s*\d+(\.\d+)*\s+', '', text)

    return cleaned_text.strip()
```